

Power BI Basics Assignment

Question 1 : What is Power BI and why is it used in businesses?

Answer:-

Power BI is a Business Intelligence and Data Visualization tool developed by Microsoft. It helps users collect, analyze, and visualize data through interactive reports and dashboards.

Why it is used in businesses:

- To analyze large amounts of data.
 - To create interactive dashboards and reports.
 - To support better decision-making.
 - To identify trends and patterns.
 - To monitor business performance in real time.
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Question 2: Name and explain the three main components of Power BI.

Answer:

1. **Power BI Desktop**
 - Used to connect, transform, and visualize data.
 - Reports and dashboards are created here.
 2. **Power BI Service**
 - Cloud-based platform used to publish and share reports.
 - Enables collaboration among team members.
 3. **Power BI Mobile**
 - Mobile application for viewing reports and dashboards.
 - Allows users to access insights anytime and anywhere.
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Question 3: Explain the Power BI workflow.

Answer:

The Power BI workflow consists of the following steps:

1. Connect to data sources.
 2. Clean and transform data using Power Query.
 3. Load data into Power BI.
 4. Create relationships between tables.
 5. Build visualizations and reports.
 6. Publish reports to Power BI Service.
 7. Share dashboards with users for analysis and decision-making.
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Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer:

1. Removing duplicate records.
2. Handling missing or null values.
3. Splitting and merging columns.
4. Changing data types (Text, Number, Date, etc.).

Other tasks include filtering rows, renaming columns, and replacing values.

Question 5: Write step-by-step instructions to load the dataset.

Answer:

1. Open Power BI Desktop.
 2. Click **Get Data** on the Home tab.
 3. Select the dataset source (Excel, CSV, SQL Server, etc.).
 4. Browse and select the dataset file.
 5. Click **Open**.
 6. Preview the data in the Navigator window.
 7. Select the required table(s).
 8. Click **Load** to import data into Power BI.
 9. If data cleaning is required, click **Transform Data** before loading.
 10. After loading, the dataset will appear in Power BI Desktop.
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Question 6: Define Data View, Report View, and Model View.

Answer:

1. Data View

- Used to view and inspect data loaded into Power BI.
- Helps verify data accuracy and structure.

2. Report View

- Used to create charts, graphs, tables, and dashboards.
- Main workspace for report development.

3. Model View

- Used to create and manage relationships between tables.
 - Helps design the data model for analysis.
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Question 7: Discuss the different data sources that Power BI supports.

Answer:

Power BI supports many data sources, including:

- Excel Files
- CSV Files
- Text Files
- SQL Server
- MySQL
- Oracle Database
- PostgreSQL
- SharePoint
- Azure Services
- Google Analytics
- Web APIs
- Online Services such as Salesforce

This flexibility allows organizations to connect data from multiple platforms into a single report.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

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= Table.RenameColumns("#Changed Type",{{"Owner_Name.1", "First Name"}, {"Owner_Name.2", "Last Name"}})

123

Bedrooms

123

Bathrooms

ABC

First Name

ABC

Last Name

123

Contact

90%

Valid

0%

Error

10%

Empty

100%

Valid

0%

Error

0%

Empty

100%

Valid

0%

Error

0%

Empty

90%

Valid

0%

Error

10%

Empty

90%

Valid

0%

Error

10%

Empty

1

850

2

Rahul

Mehta

98765432

2

720

2

Anita

Sharma

98989898

3

900

3

Ramesh

Kumar

91234567

4

880

3

Neha

Verma

99887766

5

950

3

Amit

Joshi

90909090

6

800

2

93456789

7

null

2

Suresh

Reddy

90123456

8

870

3

Divya

Iyer

95566778

9

1200

4

Manoj

Patil

98877665

10

700

2

Raj

Singh

n

11

850

2

Rahul

Mehta

98765432

12

720

2

Anita

Sharma

98989898

13

900

3

Ramesh

Kumar

91234567

14

880

3

Neha

Verma

99887766

15

950

3

Amit

Joshi

90909090

16

800

2

93456789

17

null

2

Suresh

Reddy

90123456

18

870

3

Divya

Iyer

95566778

19

1200

4

Manoj

Patil

98877665

20

700

2

Raj

Singh

n

21

850

2

Rahul

Mehta

98765432

22

720

2

Anita

Sharma

98989898

Query Settings

PROPERTIES

Name

Real_Estate_Data_100_Rows

All Properties

APPLIED STEPS

Source

Promoted Headers

Split Column by Delimiter

Changed Type

Renamed Columns